Unit 9, Lesson 2: Adding Linear Expressions with Rational Coefficients

Benchmark

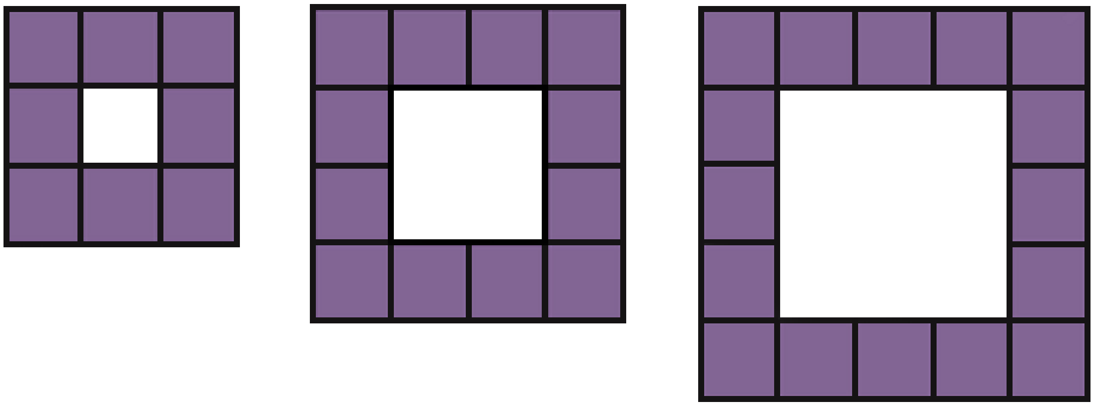
MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.

Learning Target

- ☐ I can add linear expressions with rational coefficients.

9.2.1 Warm-Up: Visual Patterns

Three figures are shown.



1. What pattern do you see?

2. Explain how many small purple squares you think will be in the next figure.

9.2.2 Exploration: Which One Doesn't Belong?

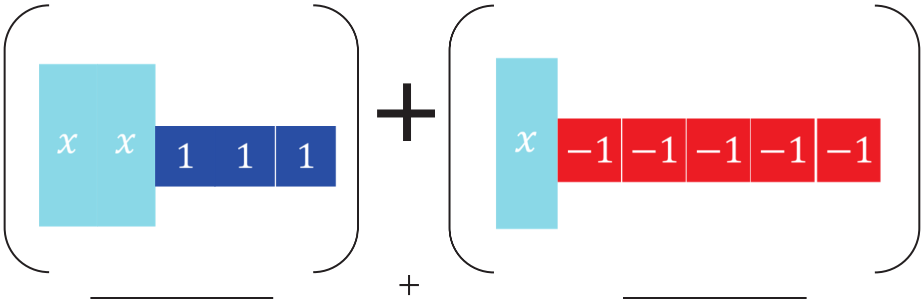
1. Which expression does not belong with the other three? Explain your reasoning.

Expression A $7 - 5x - 3$	**Expression B** $\frac{3}{2}x + 6 - 2 - 6\frac{1}{2}x$
Expression C $-5x + 7.25 - 4.25$	**Expression D** $-2x - 3x + 4$

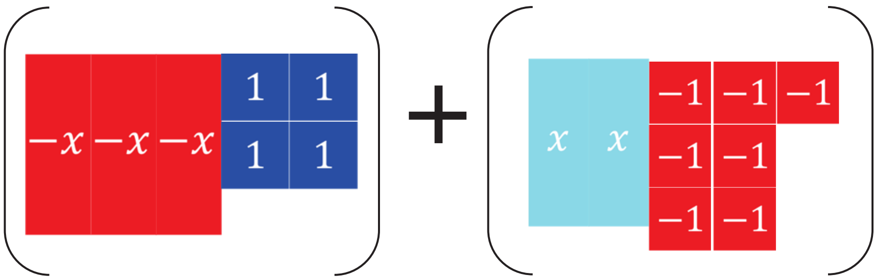
9.2.3 Collaboration: Adding Expressions with Algebra Tiles

Algebra tile representations of an expression are given. Work with your partner to find the sum by following the steps below.

1.

On the lines, write the algebraic expression for each representation.	 ______ + ______
Rearrange the tiles to group like tiles. Draw the representation.	
Eliminate any zero pairs to find the sum. Write the sum.	

2.

On the lines, write the algebraic expression for each representation.	 ______ + ______
Rearrange the tiles to group like tiles. Draw the representation.	
Eliminate any zero pairs to find the sum. Write the sum.	

9.2.4 Guided Instruction: Adding Linear Expressions

When adding two expressions, rearrange the terms so that like terms are grouped together. Then add the coefficients of the like variable terms together and add the constants. Zero pairs are eliminated in this process.

Subtracting an integer is the same as adding the opposite. For example, the expression $9 - 13$ can be rewritten as $9 + -13$.

1. Find the sum of the two expressions.

a.

Original Expression	$(2x + 6) + (5x + 4)$
Regrouped Expression: Rearrange the terms so that like terms are together.	$\_\_\_\_x + \_\_\_\_x + \_\_\_\_ + \_\_\_\_$
Simplified Expression: Add the coefficients of the like variable terms and add the constants.	$\_\_\_\_x + \_\_\_\_$

b.

Original Expression	$(-2x + 6) + (5x - 4)$
Regrouped Expression: Rearrange the terms so that like terms are together.	$\_\_\_\_x + \_\_\_\_x + \_\_\_\_ + \_\_\_\_$
Simplified Expression: Add the coefficients of the like variable terms and add the constants.	$\_\_\_\_x + \_\_\_\_$

9.2.5 Collaboration: Adding Linear Expressions

1. Work with your partner to find the sum of the two expressions in simplest form.

a.

Original Expression	$4 + (8 - 0.4x)$
Regrouped Expression	
Simplified Expression	

b.

Original Expression	$\left(3\frac{1}{2}x - \frac{3}{5}\right) + \left(2\frac{1}{4}x + \frac{1}{3}\right)$
Regrouped Expression	
Simplified Expression	

c.

Original Expression	$(-5y + 2) + (5y - 8)$
Regrouped Expression	
Simplified Expression	

d.

Original Expression	$(0.7 - 2.4x) + (4.1x - 5.32)$
Regrouped Expression	
Simplified Expression	

e.

Original Expression	$\left(\frac{3}{4}w - 5.5\right) + \left(-\frac{1}{2}w + 4\right)$
Regrouped Expression	
Simplified Expression	

9.2.6 Your Turn!

1. Find the sum of each expression using the fewest terms possible.

a. $(x + 7) + (x - 10)$	b. $(3x - 5) + (5 + 6x)$
c. $\left(1\frac{6}{7} - y\right) + \left(4y + \frac{1}{7}\right)$	d. $(4w - 6) + (-6w - 1)$
e. $(9 - 4.6x) + (-5.3x - 7)$	f. $\left(-\frac{5}{2}k - 7\right) + \left(\frac{1}{5}k + 4\right)$

9.2.7 Wrap-Up: Error Analysis

1. Your friend finds the sum of $\left(-2\frac{1}{2}x + 7.1\right) + (4x - 5.2)$ to be $-\frac{3}{2}x + 1.9$.

   Write a detailed note to your friend explaining why their answer is correct or why it is incorrect.